ALICE L. WALTON SCHOOL OF MEDICINE

# Arrhythmias: Pathophysiology and Diagnosis

Last updated January 21, 2026 · 17 min read · ScholarRx

Adapted by: Diptiman Bose, Phil Riley, Robert Donnell

Adapted from: Arrhythmias: Foundations and Frameworks

Listen to this Brick
0:00 / 28:47 1x

Learning Objectives (4) Versions

*After completing this brick, you will be able to:*

1. Describe the cardiac action potential and the conduction pathways
2. Describe the typical clinical presentation associated with arrhythmias
3. Explain the pathophysiological causes of arrhythmias based on mechanisms
4. Identify diseases/drugs that can cause arrhythmias

**CASE CONNECTION**

TF is a 56-year-old male experiencing palpitations. "It feels like a horse is galloping all around my chest," TF says. "Once, I felt like I was going to pass out. But most of the time, I just get a little dizzy. I don't know what brings them on, but I sure would like them to go away." TF has a history of hypertension but otherwise has no medical problems. He denies syncope or chest pain. His only medication is hydrochlorothiazide. On exam, TF's blood pressure is 134/82 mm Hg and his heart rate is 112 beats/min and irregularly irregular. The remainder of his examination is normal.

What is the most likely diagnosis? What treatment would you recommend? Consider your answers as you read, and we'll revisit TF at the end of the brick.

GO TO CONCLUSION ↓

## Cardiac Action Potential (LO#1)

The ventricular cardiac action potential is a coordinated electrical event that triggers contraction of cardiac muscle cells and allows the heart to pump blood efficiently. It consists of five phases (0–4), each caused by specific ion movements across the cardiac cell membrane through specialized ion channels. These phases create a controlled sequence of depolarization and repolarization that maintains normal heart rhythm and prevents premature contractions.

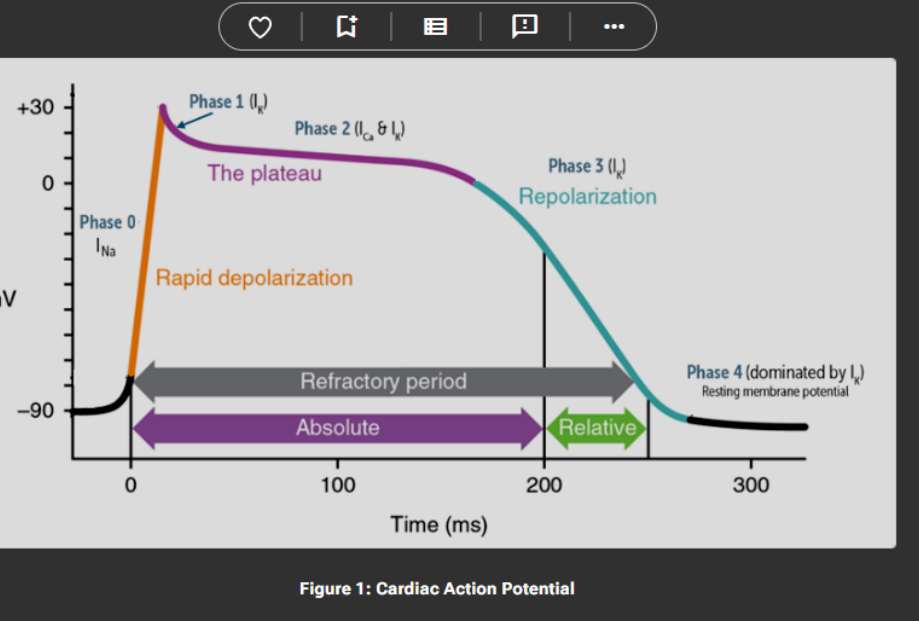


Figure 1: Cardiac Action Potential

**Phase 4 – Resting membrane potential**
The membrane potential is maintained at approximately −90 mV by continuous potassium efflux through inwardly rectifying potassium ($IK_1$) channels. Voltage-gated sodium and calcium channels remain closed, preserving ionic gradients across the membrane.

**Phase 0 – Rapid depolarization**
A depolarizing stimulus opens fast voltage-gated sodium channels, allowing a rapid influx of $Na^+$ (INa). This causes a steep rise in membrane potential from negative values to around +30 to +40 mV.

**Phase 1 – Early repolarization**
Fast sodium channels inactivate, stopping $Na^+$ influx and preventing further depolarization. At the ium channels (Ito) open, allowing $K^+$ efflux and producing the small repolarizing notch.

**Phase 2 – Plateau phase**
L-type voltage-gated calcium channels open, permitting sustained $Ca^{2+}$ influx (ICa-L) into the cell. This inward calcium current is balanced by outward potassium currents (IK), creating a prolonged plateau in membrane potential.

**Phase 3 – Repolarization**
Calcium channels close while delayed rectifier potassium channels (IKr and IKs) remain open, allowing potassium to exit the cell. This $K^+$ efflux restores the membrane potential back to its resting level, preparing the cell for the next action potential.

ACTIVE LEARNING QUESTION

**What initiates repolarization in cardiac cells?**

A. ○ Calcium influx
B. ● Potassium efflux
C. ○ Sodium influx
D. ○ Calcium efflux
E. ○ Sodium efflux

**Explanation**

The correct answer is potassium efflux (B). During repolarization, potassium channels remain open, allowing $K^+$ to exit the cell, restoring membrane potential. Calcium influx (A) occurs during the plateau phase, not repolarization. Sodium influx (C) is part of depolarization. Calcium efflux (D) does not initiate repolarization. Sodium efflux (E) is not involved in repolarization.

By prolonging the depolarized state, cardiac cells remain unresponsive to new stimuli, preventing premature or excessive contractions. This protects the heart from tetany and allows adequate time for ventricular filling between beats. It is important to recall the **refractory period for a cardiac action potential.** The refractory period is the interval after an action potential during which a cardiac cell cannot respond (absolute) or has a reduced ability to respond (relative) to a new electrical stimulus. Refractory periods protect the myocyte against arrhythmias.

## What Are Arrhythmias?

An arrhythmia is any abnormality of the heart's rate or rhythm that interrupts or replaces Normal Sinus Rhythm (NSR). Any disruption in the pattern of electrical signaling that occurs during the cardiac cycle will result in a heartbeat that is too fast, too slow, or otherwise irregular. Arrhythmias are common, affecting about 0.5% of the population annually. They present with a range of symptoms, f hreatening, causing sudden cardiac death.

## How Do Patients With Arrhythmias Present? (LO#2)

Arrhythmias present in a variety of ways. Patients can be asymptomatic or have symptoms ranging from the sensation of an extra beat, to spontaneous dizziness to cardiac arrest. The clinical presentation depends on the **regularity** of the beat, **rate** of the rhythm if too fast or too slow and the **rate of speeding or slowing** from the underlying normal rhythm.

Fast heart rates most likely lead to perceptible **palpitations** or sometimes symptoms of **ischemic heart disease** like chest pain, nausea, and diaphoresis, because the rapid heartbeat increases oxygen consumption of the myocardium. Moreover, if the heart rate is "ultra" fast or there is sufficient underlying cardiovascular disease to impair the patient's compensatory mechanisms, they can lead to syncope and even cardiac arrest. Further discussion on this is found later in this brick.

CLINICAL CORRELATION

**Sinus tachycardia** at rest is usually secondary to an underlying disease process like infection, anemia, pulmonary embolism, heart failure or hyperthyroidism (to name a few). We call the heart rate in this situation a "compensatory mechanism". In this situation we describe Sinus Tachycardia as being "secondary" to another medical illness. ***Resting Sinus Tachycardia is never normal.***

In addition, slow heart rates can also cause **dizziness** or **syncope** because the slow heart rate can decrease cardiac output, limiting oxygen delivery to the brain. Extreme degrees of bradycardia can also occur, leading to syncope or even cardiac arrest.

How do arrhythmias present?

## What Is the Pathophysiology of Arrhythmias? (LO#3)

Let's look at some of the underlying causes and how these can disrupt normal conduction.

### Classification of arrhythmias

There are at least thr n, these being by **mechanism, by rate and by site of origin.**

Arrhythmias occur because of one of five primary underlying mechanisms: enhanced automaticity (tachycardias), decreased automaticity (bradycardias), triggered activity (tachycardias), reentry (tachycardias), or conduction block (bradycardias).

### Classification by mechanism

### Automaticity

**Automaticity** refers to the ability of a myocardial cell to generate its own action potential spontaneously. This property is normal in the sinoatrial node and is responsible for the normal cardiac rhythm. The property of automaticity can also be a mechanism of arrhythmia. Many sites in the conduction system *below* the level of the sinoatrial node also posses the property of automaticity. These sites "take over" to make the heart beat if the SA Node fails. The intrinsic sites and rates of the various cardiac tissues are as follows: Sinus node (60-80 bpm), atria (40-50 bpm), AV junction (30-40 bpm) and ventricles (15-30 bpm). (Figure 2)

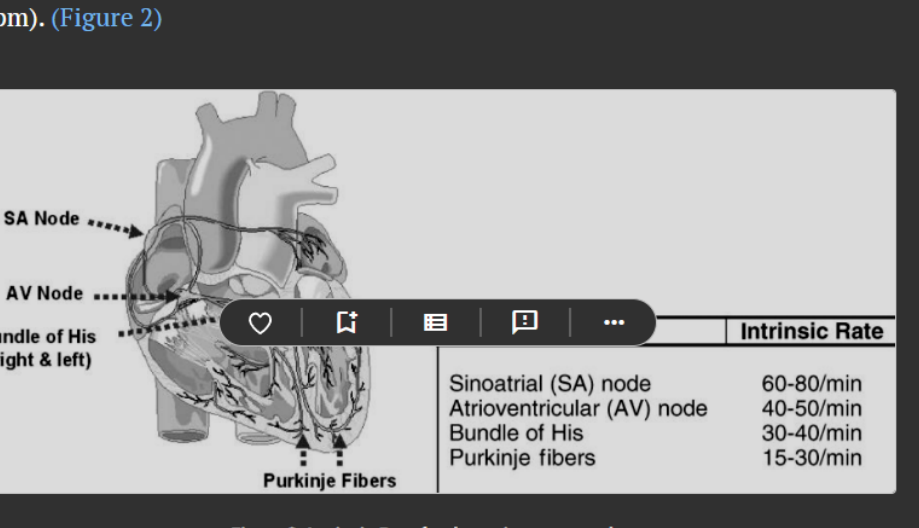

SA Node
AV Node
Bundle of His (right & left)
Purkinje Fibers
Intrinsic Rate
Sinoatrial (SA) node 60-80/min
Atrioventricular (AV) node 40-50/min
Bundle of His 30-40/min
Purkinje fibers 15-30/min

Figure 2: Intrinsic Rate for the various pacemakers

Again, these sites can serve as rescue (aka default) pacemakers if the sinoatrial node fails or if AV block prevents sinus impulses from getting through. If abnormally excited, their automaticity abnormally increases resulting in usurpation of SA Nodal function thus producing a tachycardia.

#### Increased Automaticity

*Enhanced normal automaticity* can occur in both muscle and conduction tissues in the heart. This occurs when the cells are responding to different stimuli. Examples of such stimuli include activation of Sympathetic Nervous System, inhibition of Parasympathetic Nervous System, ischemia, and drugs that are used in the intensive care unit (norepinephrine, dopamine and dobutamine to name a few)

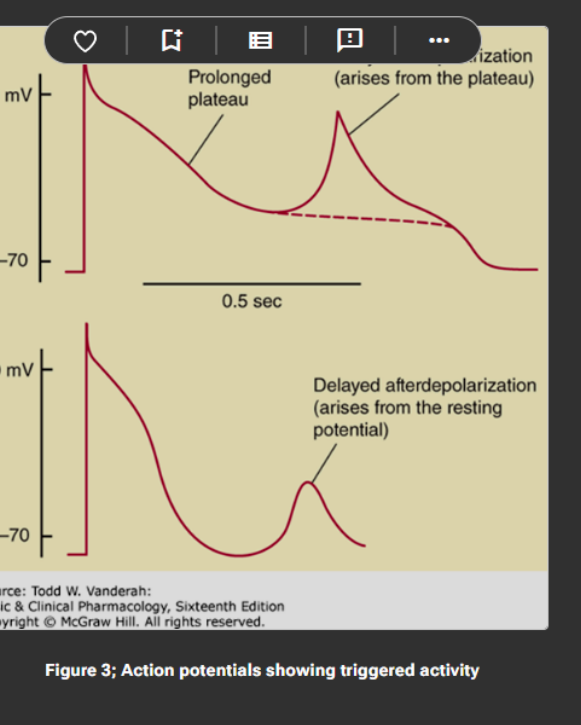
#### Decreased Automaticity

Decreased automaticity refers to a **decreased rate of action potential generation** by the SA node, resulting in a slow heart rate. The primary example is sinus bradycardia, in which the SA node continues to set the pace of the heart at a reduced rate of less than 60 beats/min. Causes can be physiologic (eg, sleep or in aerobically healthy athletes - mediated by enhanced vagal nerve tone), medication-in nnel blockers), or pathologic (eg, ischemic heart disease, cardiomyopathy, and hypothyroidism). Should this occur one of the rescue pacemakers referenced above may take over by default.

What is sinus bradycardia?

### Triggered Activity

In this mechanism, a normal action potential induces a spontaneous abnormal action potential during or immediately after it. This abnormal action potential is referred to as "after-depolarization," and it may interrupt phase 2, 3, or 4 of the action potentials. Based on the etiology, there are two major forms of afterdepolarizations: early afterdepolarizations (EADs) and delayed afterdepolarizations (DADs) (Figure 3). Table 1 distinguishes EADs from DADs.



Source: Todd W. Vanderah: Basic & Clinical Pharmacology, Sixteenth Edition Copyright © McGraw Hill. All rights reserved.

Figure 3: Action potentials showing triggered activity

The table 1 below elaborates on EADs and DADs. For help in understanding, please check out Dr. Strong's YouTube video linked below.

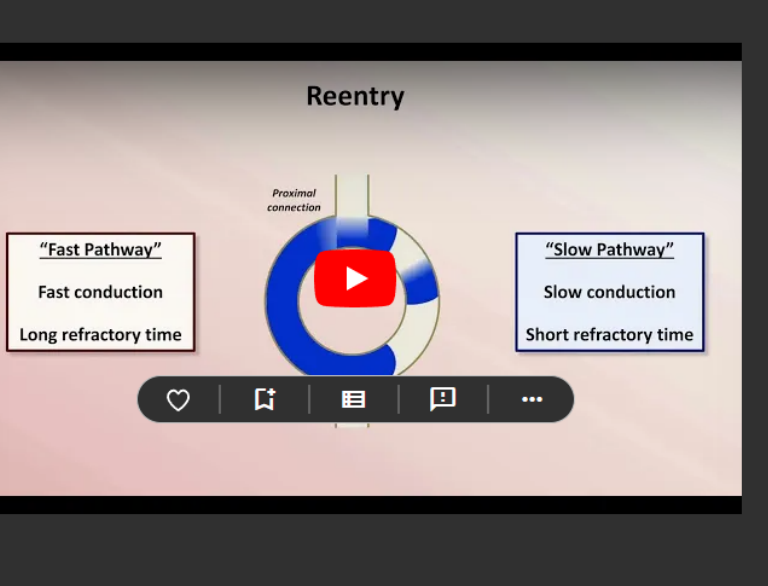
Table1

| | EAD | DAD |
|---|---|---|
| Mechanism | Prolonging action potential and action potential duration | Increased intracellular Ca overload |
| Etiologies | Hypokalemia, QT prolongation (drug-induced, QT syndrome) | Digoxin toxicity, adrenergic excess, ischemia |

### Reentry

Reentry refers to **continuous propagation** of an electrical impulse traveling in a circuitous path. This results in reactivation of the original site that generated the electrical impulse, thus continuing the cycle in a loop and causing tachycardia. This mechanism is common and can be seen in both supraventricular and ventricular tachycardias.

Reentry tachycardias are self-perpetuating electrical short circuits in the heart. There are a number of ways this can happen, but all reentrant circuits have three common features: two adjacent electrical pathways with different conduction velocities (one fast and one slow), the short circuit starts with a premature action potential (frequently called an extrasystole), and a temporary conduction block in one of the pathways. When these three conditions are all met, a leading edge and trailing tail of depolarization can then self-perpetuate in rapid fashion causing tachycardia. The following video does a good job describing this further.



A common example, as illustrated in video above, is AV Nodal Reentrant Tachycardia (AVNRT), in which a premature beat in the presence of a "slow" pathway causes electrical conduction to move from the ventricles back to the atria, rather than their normal pattern of propagating forward into the ventricles (Figure 4).

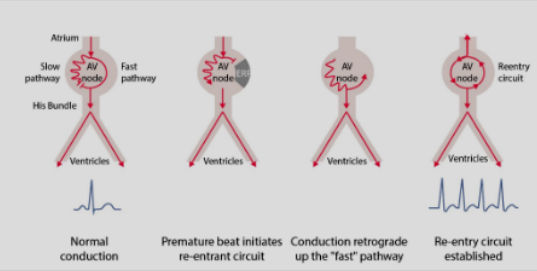

Atrium
Slow pathway
AV node
Fast pathway
His Bundle
Ventricles
Reentry circuit
Normal conduction
Premature beat initiates re-entrant circuit
Conduction retrograde up the "fast" pathway
Re-entry circuit established

Figure 4 Development of atrioventricular node reentrant tachycardia

The premature beat (Figure 2 ) is describing when a Premature Atrial Contraction occurs at just the right time that a temporary conduction block is present in one of the pathways) finds the fast pathway unavailable because its refractory period has not expired. In this particular case the refractory period is absolute, that is it has "all or none" conduction. Once the tissue recovers from the refractory period, fast conduction resumes. Because the fast pathway is blocked early in th n only travel down the slow pathway. By the time it reaches the distal AV node, the fast pathway has recovered, allowing the impulse to travel back *up* (retrograde) to the atria. You can see from Dr Strong's video and from the figure how this allows it to go round and round in an endless loop.

In addition to AVNRT, many other arrhythmias are based on reentry including atrial flutter and ventricular tachycardia. AVNRT is an example of *micro* reentry in which the circuit is < 2 cm in diameter.

Another reentrant arrhythmia is Wolff Parkinson White syndrome (WPW). This occurs in 1:5000 people. This is a rare but well known congenital heart defect in which the AV valve ring insulators are incompletely formed, allowing a strip of myocardium to connect the atria and the ventricles *directly*, thus bypassing the AV junction. The conduction velocity of this abnormal strip of myocardium is faster than that of the AV node though not as fast as that found in the Purkinje fibers in the bundle branch system.

Given that, what might be the consequences clinically? In WPW with sinus rhythm, the morphology of the QRS is often changed. What might happen if a patient goes into atrial fibrillation or flutter? Finally, the availability of an abnormal strip of myocardium connecting atria and ventricles (aka the accessory pathway or, eponymously, the Kent bundle) might enable a reentry circuit. (Figure 5)

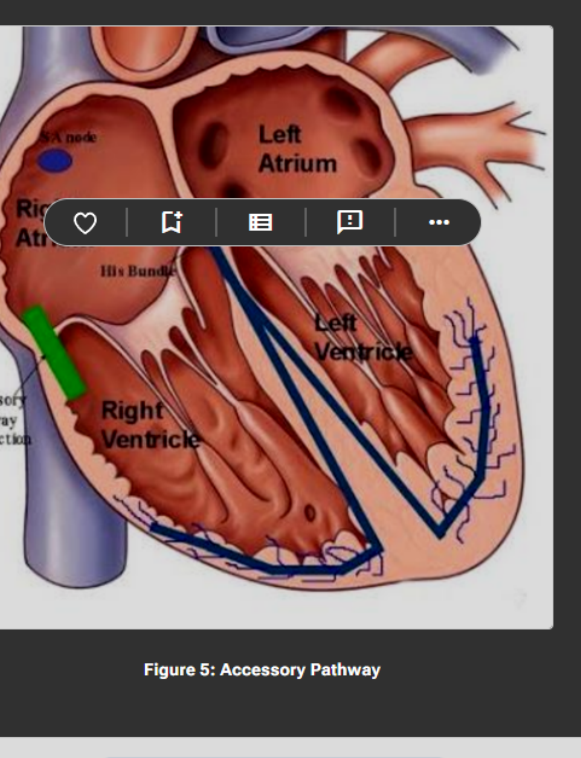


Figure 5: Accessory Pathway

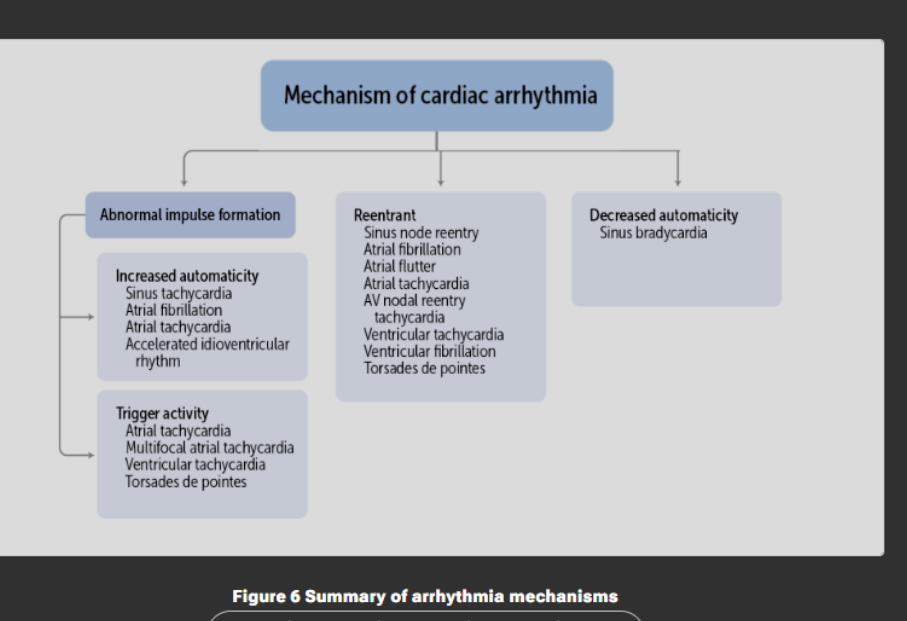


Figure 6 Summary of arrhythmia mechanisms

### Classification based on rate

Another means of classification is the simple and binary division, based on the rate into tachycardia and bradycardia. (There are further subdivisions within this scheme as discussed below).

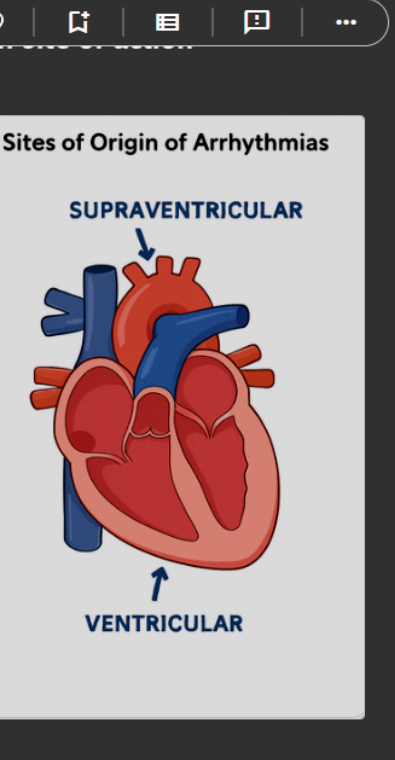
#### Tachycardia

Tachycardia may be an important compensatory mechanism in conditions of increased metabolic demand such as acute illness or exertion. This benefit is by virtue of the effect of increasing cardiac output. Remember the cardiac output equation CO= HR * SV. However, with tachycardia, a point of diminishing return may be reached with very high rates where stroke volume begins to fall (due to decreased diastolic filling time) thereby overcoming the benefits of the increased heart rate and tipping the cardiac output product downward. In addition, tachycardia increases myocardial oxygen consumption, which could lead to myocardial ischemia in patients with coronary artery disease. Some tachycardias can become "ultra fast" or degenerate into ventricular fibrillation (VF) resulting in syncope or cardiac arrest.

#### Bradycardia

In bradycardia the detrimental effect on cardiac output is more intuitive. While slower heart rates, at least within the normal range, may be beneficial in some situations, heart rates below the normal range in most cases are associated with decreased cardiac output and the attendant manifestations of dizziness, syncope or even cardiac arrest at extreme levels of bradycardia.

### Classification base



Arrhythmia mechanism can be classification based on their site of origin. They are subdivided into supraventricular and ventricular categories, referring to its site of origin above or in the ventricles, respectively.

More specifically, **supraventricular arrhythmias** arise at or above the bundle of His. Ventricular arrhythmias arise below that level. Supraventricular tachycardias tend to have a normal appearing QRS whereas ventricular arrhythmias tend to have a wide and abnormal appearing QRS. (Figure 7)

Clinically, **ventricula** red worse than supraventricular tachycardias. Why? Ventricular tachycardia is, for one thing, more common in patients with structural heart disease and therefore reduced ability to compensate hemodynamically. Also, consider the hemodynamic consequences of an abnormal activation sequence in the ventricles.

CLINICAL CORRELATION

Some VTs can become"ultra fast" and may degenerate into ventricular fibrillation resulting in cardiac arrest.

To elaborate further on the subtypes of *bradycardia*, again based on the site of the disturbance, sinoatrial node (SAN) dysfunction is excessive slowing, or even stoppage, of the sinoatrial node, the principal pacemaker of the heart.

CLINICAL CORRELATION

*Ectopic beats* (or "extrasystoles") are beats that are "out of place" in timing and sometimes morphology. They can be designated by their site of origin (ventricular, junctional, atrial ) as well as by their timing (premature or escape).

What are the primary mechanisms of arrhythmias?

### Investigation of Arrhythmias

Investigating the underlying medical cause of the arrhythmia should take place as soon as possible once the patient is stabilized.

Testing (and follow-up testing) might include:

- **Serum lab studies** that identify extra-cardiac causes, such as thyroid function or electrolyte abnormalities such as in serum potassium or calcium.
- **Electrocardiographic data** as discussed above.
- **Echocardiography** is often indicated, as it can identify the structural and functional abnormalities that may be causing an arrhythmia, like a cardiomyopathy, valve disease, or a cardiac tumor.
- **Evaluation for ischemia** (which can take many forms such as coronary angiography or non invasive evaluation for ischemia such as stress testing) is important in many patients with arrhythmias.

## What are some diseases/drugs that cause arrhythmias? (LO#4)

Some of the many causes of arrhythmias are:

- Cardiovascular disease: cardiomyopathy, ventricular hypertrophy, valvular disease, ischemic heart disease, hypertension, chest trauma
- Systemic diseases: chronic obstructive pulmonary disease, obstructive sleep apnea, diabetes mellitus, hyperthyroidism, hypoxemia (any cause)
- Electrolyte disturbances: hyperkalemia, hypokalemia, hypercalcemia, hypocalcemia, hypermagnesemia, hypomagnesemia
- Medications: digoxin, β-blockers, calcium-channel blockers, lithium, opioids, anesthetics, Illicit drugs: cocaine or amphetamines
- Surgery: postoperative, prior heart surgery

**CASE CONNECTION** BACK TO INTRODUCTION ↑

Thinking back to TF, what is the most likely diagnosis? What treatment would you recommend?

Based on TF's symptoms and the irregularly irregular rhythm on physical examination, you suspect atrial fibrillation, a common supraventricular arrhythmia. ECG findings confirm the diagnosis. TF starts on a β-blocker to control his heart rate and anticoagulant therapy to prevent embolic strokes.

## Summary

- **Cardiac Action P** al consists of five phases driven by specific ion movements, ensuring coordinated heart contractions and rhythm stability.
- **Refractory Period:**The refractory period protects cardiac cells from premature activation, reducing the risk of arrhythmias by limiting responsiveness after depolarization.
- **Arrhythmia:**Arrhythmias are disturbances in heart rate or rhythm, presenting with symptoms ranging from mild palpitations to life-threatening cardiac arrest.
- Symptoms of arrhythmias vary by rate and regularity, including palpitations, dizziness, syncope, and ischemic chest pain.
- **Classification by Mechanism:**Arrhythmias are classified by underlying mechanisms such as automaticity changes, triggered activity, reentry.
- **Automaticity:**Automaticity refers to the heart's ability to generate spontaneous action potentials, with abnormalities leading to arrhythmias.
- **Triggered Activity:**Triggered activity involves abnormal afterdepolarizations, often caused by electrolyte imbalances or drug toxicity, leading to arrhythmias.
- **Reentry:**Reentry describes a self-perpetuating electrical circuit in the heart, commonly responsible for tachyarrhythmias like AVNRT and WPW.
- **Drugs/Disease Causing Arrhythmias:** Various diseases, electrolyte disturbances, and medications can precipitate arrhythmias, highlighting the need for thorough evaluation.

## Review Questions

**1. What initiates repolarization in cardiac cells?**

A. ○ Calcium influx
B. ● Potassium efflux
C. ○ Sodium influx
D. ○ Calcium efflux
E. ○ Sodium efflux

**Hide Explanation**

The correct answer is potassium efflux (B). During repolarization, potassium channels remain open, allowing $K^+$ to exit the cell, restoring membrane potential. Calcium influx (A) occurs during the plateau phase, not repolarization. Sodium influx (C) is part of depolarization. Calcium efflux (D) does not initiate repolarization. Sodium efflux (E) is not involved in repolarization.

**2. A 45-year-old woman presents with episodes of dizziness and palpitations. She reports that her symptoms often occur when she is anxious or after drinking coffee. Her ECG shows a heart rate of 120 beats per minute with normal QRS complexes. She has no history of heart disease. Laboratory tests are unremarkable. The physician suspects an arrhythmia related to abnormal pacemaker activity. Which of the following mechanisms most likely explains her arrhythmia?**

A. ● Enhance
B. ○ Decreased automaticity in the atrioventricular node
C. ○ Reentry circuit in the ventricles
D. ○ Conduction block in the bundle branches
E. ○ Prolonged refractory period in the atria

**Hide Explanation**

The correct answer is (A): Enhanced automaticity refers to an increased rate of spontaneous action potential generation, often triggered by sympathetic stimulation (e.g., anxiety, caffeine). This can lead to tachyarrhythmias with normal QRS complexes. (B) is incorrect: Decreased automaticity would cause bradycardia, not tachycardia. (C )is incorrect: Reentry circuits typically cause arrhythmias like atrial flutter or ventricular tachycardia, often with abnormal QRS morphology. (D) is incorrect: Conduction block leads to heart block or bradyarrhythmias. (E )is incorrect: Prolonged refractory period prevents premature beats, not tachyarrhythmias due to increased automaticity.

**3. Which arrhythmia is most likely to present with a wide and abnormal QRS complex on ECG?**

A. ○ Supraventricular tachycardia
B. ○ Sinus tac
C. ● Ventricular tachycardia
D. ○ Atrial fibrillation

**Hide Explanation**

The correct answer is ventricular tachycardia (C). Ventricular tachycardia originates below the bundle of His, causing abnormal ventricular activation and a wide QRS complex. The other rhythms (A, B, and D) Supraventricular tachycardia, sinus tachycardia, atrial fibrillation, typically have narrow QRS complexes unless there is aberrant conduction.

**4. A 60-year-old man with a history of hypertension and heart failure presents to the clinic complaining of palpitations and intermittent dizziness. He recently started taking digoxin for his heart failure. On examination, his pulse is irregular and ECG reveals frequent premature ventricular contractions. Laboratory tests show mild hypokalemia. Which of the following is most likely cause of the patient's arrhythmia?**

A. ○ Hyperthyroidism
B. ○ β-blocker therapy
C. ○ Hypocalcemia
D. ○ Chronic o
E. ● Digoxin toxicity

**Hide Explanation**

The correct answer is (E): Digoxin toxicity is a well-known cause of arrhythmias, especially in the setting of hypokalemia, which increases the risk of digoxin-induced arrhythmias such as premature ventricular contractions and other ectopic beats. (C) is incorrect: Hyperthyroidism can cause arrhythmias (e.g., atrial fibrillation), but there is no evidence of thyroid disease in this vignette. B is incorrect: β-blockers typically slow the heart rate and are more likely to cause bradyarrhythmias rather than ectopic ventricular beats. (C) is incorrect: Hypocalcemia can affect cardiac conduction, but is less commonly associated with arrhythmias than digoxin toxicity in this context. (D) is incorrect: COPD can predispose to arrhythmias, but the temporal relationship with digoxin initiation and hypokalemia makes digoxin toxicity the most likely cause.

**How would you rate this Brick?**

👍 👎

## References

- Issa ZF, Miller JM, *Electrophysiology*. Elsevier; 2019.
- Katritsis DG, Morady F. *Clinical Cardiac Electrophysiology*. Elsevier; 2022.
- https://www.sciencedirect.com/science/article/pii/S1880427615001647
- https://www.uptodate.com/contents/arrhythmia-management-for-the-primary-care-clinician?search=cardiac%20arrythmias&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- https://www.uptodate.com/contents/reentry-and-the-development-of-cardiac-arrhythmias?search=cardiac%20arrythmias&source=search_result&selectedTitle=4~150&usage_type=default&display_rank=4

**Review Learning Objectives**

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the cardiac action potential and the conduction pathways~~
02 ☑ ~~Describe the typical clinical presentation associated with arrhythmias~~
03 ☑ ~~Explain the pathophysiological causes of arrhythmias based on mechanisms, rate, and site of origin.~~
04 ☑ ~~Identify~~

01647
- https://www.uptodate.com/contents/arrhythmia-management-for-the-primary-care-clinician?search=cardiac%20arrythmias&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- https://www.uptodate.com/contents/reentry-and-the-development-of-cardiac-arrhythmias?search=cardiac%20arrythmias&source=search_result&selectedTitle=4~150&usage_type=default&display_rank=4

**Review Learning Objectives**

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the cardiac action potential and the conduction pathways~~
02 ☑ ~~Describe the typical clinical presentation associated with arrhythmias~~
03 ☑ ~~Explain the pathophysiological causes of arrhythmias based on mechanisms, rate, and site of origin.~~
04 ☑ ~~Identify diseases/drugs that can cause arrhythmias~~

Objective Confidence: 4/4